

→ Data :

Known facts that can be recorded and have an implicit meaning

Database :

Collection of related data

DBMS :

DB + Software

Mini-world :

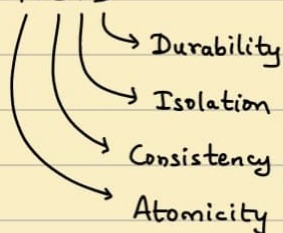
Some part of the real world about which data is stored in DB.

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Triggers and Stored Procedure → Not part of syllabus

Keywords : Entity , Attribute , Relations

- ACID



- Constraints , Operations , Datatypes

Database schema → Description | Catalogue

State → Snapshot

- Three - State Architecture

Two - tier Client - Server architecture

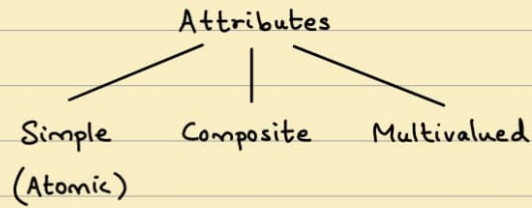
- DDL ← Data creation

DML ← Query Language

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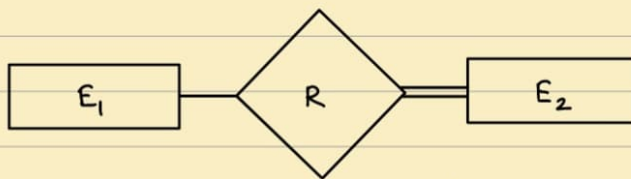
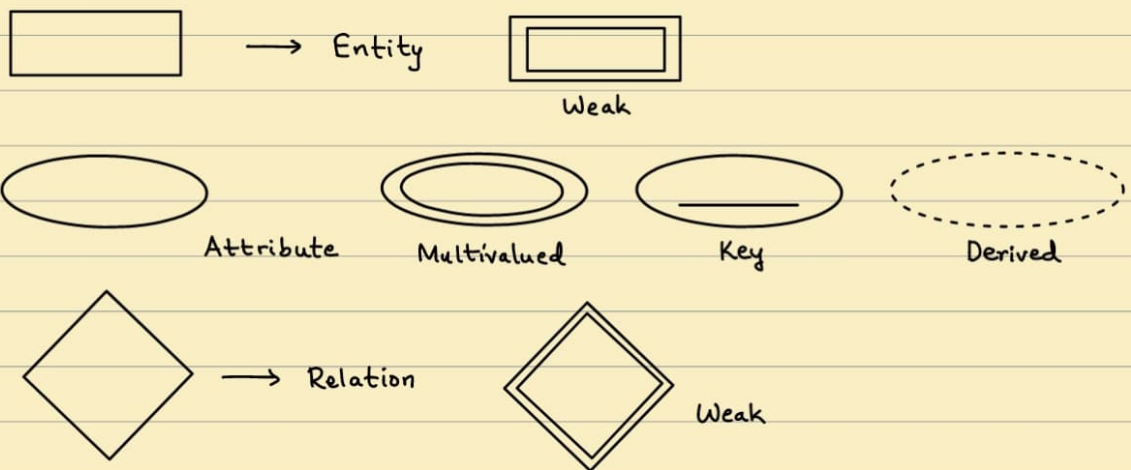
→ ER Model

- Database Designer : doesn't care how the data is captured.
 - Entity : Basic concept of ER Model (Row)
- Attributes : Properties of entity (column)



Key Attribute → Each entity has a unique value

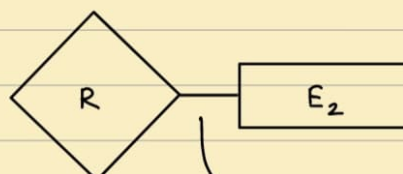
Representation : Entity. Attribute



→ Total participation of E_2 in R



→ Cardinality Ratio → 1:N for $E_1:E_2$ in R



→ (min-max) → Structural constraint

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Relation types

Constraints

Recursive relations

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- Weak Entity → Depends on strong Entity



- Attributes of Relationship types

- Cardinality ratio

Participation constraint

- min-max notation

If you query often → Derived Attribute X
Relationship type ✓

∴ Fetching is much faster

→ Data Models:

(1) ER Models

(2) UML Class diagram

Entities ↔ Classes

Relationships ↔ Lines b/w classes

→ DBMS:

Scripting Language

Embedded Approach

- Interface

- Utilities → Functions

- Cambridge Analytica - 3500 features stored about you

- Centralised DBMS Architecture

Client - Server Architecture → 3-Tier

→ Relational Model :

- Relation → Set of rows, i.e. tuple
Attribute → Column
→ Ordered set of values (1 tuple : 1 Row)
- Key : Uniquely identifies a row in a table
→ Can be set of attributes (or) attribute

- Schema : Table definition

$R(A_1, A_2, \dots, A_n)$
→ Attributes
→ name of the relation

- Domain :

- All possible column values
- Datatype constraint
Format constraint
- Developed from Attribute name

- Relation State :

$r(R)$ → Subset of Relation R

$$r(R) = \{t_1, t_2, \dots, t_n\}$$

$$t_i = \langle v_1, v_2, \dots, v_n \rangle$$

v_j → element of domain of A_j

$$r(R) \subset \text{dom}(A_1) \times \text{dom}(A_2) \times \dots \times \text{dom}(A_n)$$

The tuples are not considered to be ordered.

The attributes are ordered. ← Self-Describing

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- Value : Atomic

value v_i of attribute A_i for tuple $t \leftarrow t[A_i] = v_i$

- Constraints :

- (1) Inherent (Implicit) : No duplicates
- (2) Schema-based (Explicit) : DB design
- (3) Application-based (Semantic) : By programmers

- Super Key : Attribute / set of attributes that uniquely identify all attributes in a relation
i.e. for any 2 distinct tuples t_1 & t_2 in $r(R)$: $t_1[SK] = t_2[SK]$
- Minimal Superkey $\rightarrow$ Key / Candidate Key
- Primary Key $\rightarrow$ Smallest of the candidate Keys
 $\rightarrow$ Cannot be NULL
- Foreign Key $\rightarrow$ Referential Integrity constraint

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$\rightarrow$ Keys

$\rightarrow$ Integrity Violation

- Domain constraint
- Key constraint
- Referential Integrity constraint
- Entity Integrity constraint

ON DELETE $\left\{ \begin{array}{l} \text{RESTRICT} \\ \text{SET NULL} \\ \text{CASCADE} \end{array} \right.$

$\rightarrow$ SQL :

AUTHORIZATION — Owner of Schema

Catalogue $\rightarrow$ Collection of Multiple Schemas

SELECT Fname, Mname, Lname

FROM EMPLOYEE

WHERE Dno = $\left(\begin{array}{l} \text{SELECT Dnumber} \\ \text{FROM DEPARTMENT} \\ \text{WHERE Dname = "RESEARCH"} \end{array} \right)$

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AS
LIKE
BETWEEN
ORDER BY



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UPDATE SYNTAX

DELETE SYNTAX

JOINS

- Natural JOIN
- Outer JOIN (left & Right)
- Inner JOIN

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→ Functional Dependencies :

- No normalization → Redundant information (Storage waste)
 ↳ INSERT, UPDATE & DELETE Anomalies

• Guidelines :

- 1) Only keep relevant information
- 2) Avoid Anomalies
- 3) Reduce NULLs
- 4) Avoid Spurious tuples on JOIN

$$(r(R) : (t_1[X] = t_2[X] \Rightarrow t_1[Y] = t_2[Y])) \Rightarrow "X \rightarrow Y"$$

• Normalization :

1st Normal Form

2nd Normal Form

3rd Normal Form

BCNF (3.5 Normal Form) → Keys + Functional Dependencies

4th Normal Form → Keys + Multivalued Dependencies

5th Normal Form → Keys + Join Dependencies

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→ Anamolies :

INSERT Anamololy

UPDATE Anamololy

→ Prime Attributes : Member of some candidate Keys

→ Normalisation :

• 1 NF :

No Multivalued Attributes

No Composite Attributes

No Nested Relations , i.e. only Atomic attributes are allowed

• 2 NF :

1st Normal & No Partial Dependency i.e. Prime $\rightarrow$ Non-Prime X

All non-Prime attributes should be fully functional dependent on C.K/S.K.

Partial Dependency :

$X \rightarrow Y$

X : Proper subset of CK , i.e. Not including the CK

Y : Non-Prime Attribute

Break FDs into independent relations.

$$\left. \begin{array}{l} \underline{A} \rightarrow B \\ \underline{C} \rightarrow D \\ \underline{C} \rightarrow E \end{array} \right\} \Rightarrow \begin{array}{ll} \underline{A} \rightarrow B & \underline{C} \rightarrow D \\ & \underline{C} \rightarrow E \end{array}$$

• 3 NF :

2nd Normal + No transitive dependency

Transitive Dependency : Non-Prime $\rightarrow$ Non-Prime

$X \rightarrow Y$

$\forall F.D \left\{ \begin{array}{l} X : \text{LHS must be a CK or SK} \\ Y : \text{RHS is a Prime Attribute} \end{array} \right. \text{ [OR]}$

$$\left. \begin{array}{l} \underline{A} \rightarrow B \\ \underline{A} \rightarrow C \\ \underline{A} \rightarrow D \\ D \rightarrow E \end{array} \right\} \Rightarrow \begin{array}{ll} \underline{A} \rightarrow B & \underline{D} \rightarrow E \\ \underline{A} \rightarrow C & \\ \underline{A} \rightarrow D & \end{array}$$

Q) LOTS (A, B, C, D, E, F)

BC $\rightarrow$ Candidate Key

A $\rightarrow$ Primary Key

$A \rightarrow \{B, C, D, E, F\}$

$\{B, C\} \rightarrow \{A, D, E, F\}$

$D \rightarrow E$

$B \rightarrow F$

2NF : $B \rightarrow F$ (Violation)

A $\rightarrow \{B, C, D, E\}$

$\{B, C\}$ $\rightarrow \{A, D, E\}$

$D \rightarrow E$

B $\rightarrow F$

3NF : $D \rightarrow E$ Violation

A $\rightarrow \{B, C, D\}$

$\{B, C\}$ $\rightarrow \{A, D\}$

B $\rightarrow F$

D $\rightarrow E$

• BCNF : (3.5 NF)

3rd Normal form &

$X \rightarrow Y$

$\forall F.D, X$: LHS must be a CK or SK

i.e. $X \rightarrow$ Superkey

e.g. (A, B, C, D)

$A \rightarrow \{B, C, D\}$

$\{B, C\} \rightarrow \{A, D\}$

$D \rightarrow A$

} $\Rightarrow$

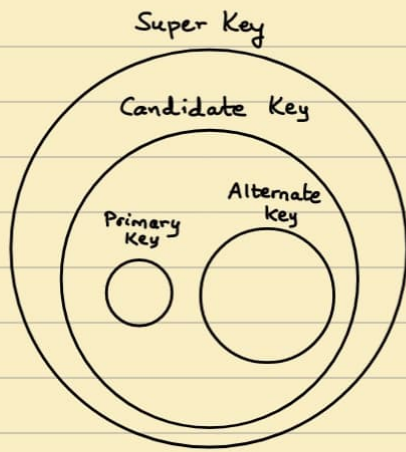
A $\rightarrow \{C, D\}$

D $\rightarrow B$

$\rightarrow$ Trivial FD : $X \subset Y$

Non-Trivial FD : $X \cap Y = \emptyset$

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Candidate Key : Minimal Superkey

Primary Key : Chosen Candidate Key

→ Nested Queries

- IN Operator
- ANY, SOME, ALL Operators

→ GROUP BY - HAVING :

COUNT, SUM, MAX, MIN, AVG ← Aggregate functions

→ IS NULL

IS NOT NULL

• EXISTS

NOT EXISTS

→ SELECT

FROM

WHERE

GROUP BY

HAVING

ORDER BY

Syntax

→ VIEW :

Virtual Table

- DROP

Thank You